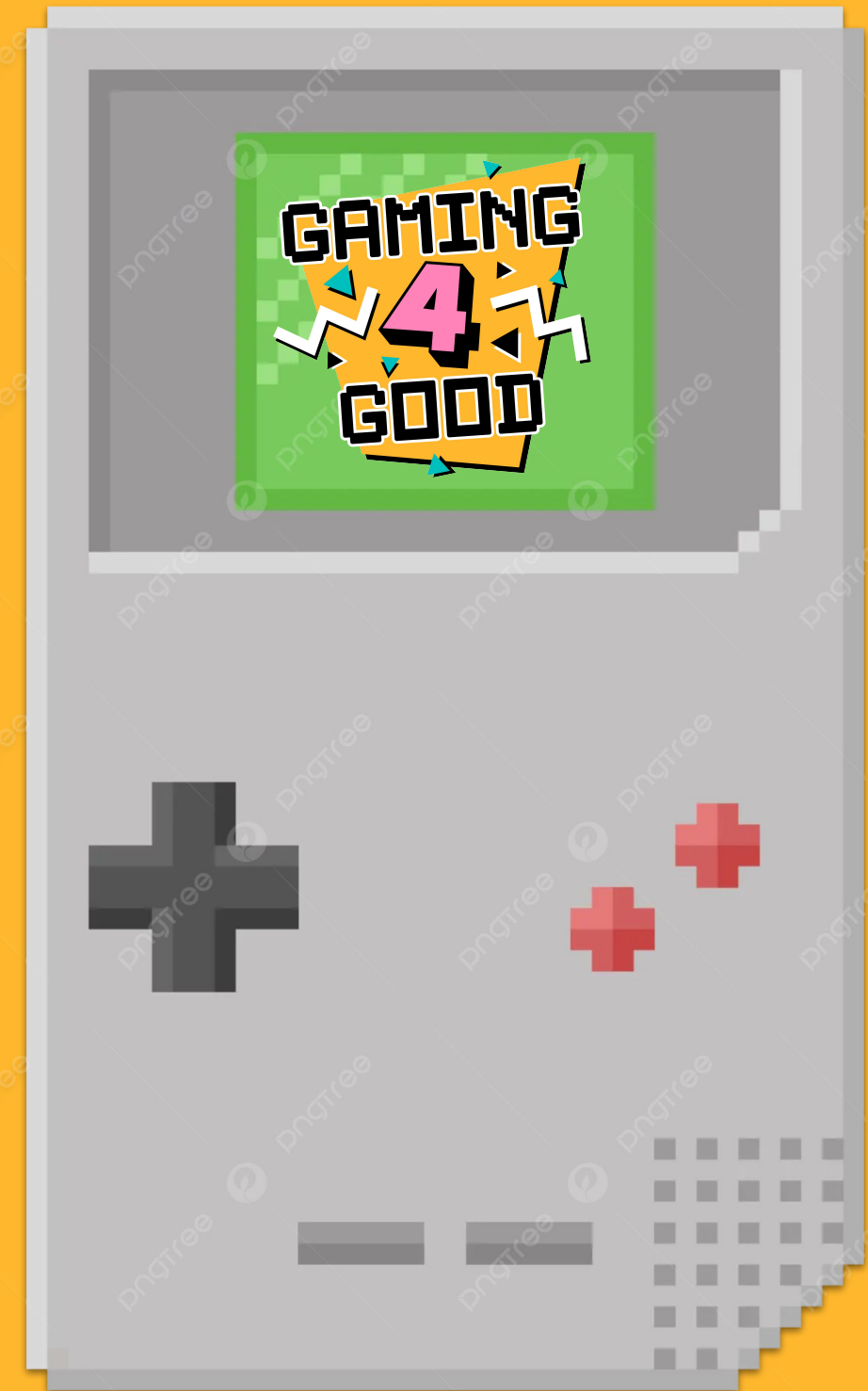


Reimagining Computational Thinking and STEM Learning with Neurodivergent Youth Through Rightful Presence and Pedagogies of Play

Ren Mendoza, Noah Glaser, Lucas Jensen, Matthew Schmidt, Amanda Olsen, Tanya Churaman, Yupei Duan, & David Ogundairo



GAMING4GOOD

OUR PROGRAM

- Pilot of a community-based, game jam program implemented across multiple informal settings
- Designed for neurodivergent middle school youth
- Youth designed original video games using Nintendo's Game Builder Garage (GBG)
- Focus on CT practices and skills as well as STEM interest and identity



GAME JAMS

- Game design supports development of computational thinking through creating playable artifacts (Kafai & Burke, 2015)
- Involves core CT practices like problem-solving and debugging (Kazimoglu et al., 2012)
- Provides accessible entry points for learning programming and CT (Akcaoglu & Kale, 2016)
- Emphasizes design-based learning through active creation (Jun et al., 2017)
- Participatory design practices create opportunities for youth to explore STEM interest and identity (Kafai & Burke, 2015)

SITES AND PARTICIPANTS

IMPLEMENTATION

Site	Site Description	Program Duration	<i>n</i> youth
Midwest			
1	Summer School Program	Two-hours, twice weekly for four weeks	10
2	Library-Based Summer Camp	Two-hours, twice weekly for four weeks	11
South			
3	STEM School Summer Camp I	Four hours a day for 5 days	8
4	STEM School Summer Camp II	Four hours a day for 5 days	6
5	Library-Based Summer Camp	One day pilot of 4 hours	6

THE LEARNING ENVIRONMENT



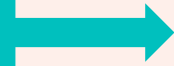
SUPPORTS



- Visual modeling; scaffolding
- Verbal + drawn explanations
- Structured schedules + flexibility check-ins

Instructional supports for engagement and access

CHOICE



- Game themes
- Story design
- Characters

Supporting student-driven interests in project work

ACCESSIBILITY

- Individual + flex spaces
- Sensory supports and adaptive space
 - Adjustable lighting
 - Flexible seating
 - Breaks + snacks

Emphasis on comfort and accessibility



GAME JAM STRUCTURE

- 🌿 Platform: Nintendo Game Builder Garage (GBG)
- 🌿 8 sequenced modules aligned with Jenson et al. (2024)
- 🌿 Focuses on: Core CT concepts & Game design practices
- 🌿 Prototyping, debugging, playtesting, exploration
- 🌿 Culminates in Original student-created games

GAME JAM

CURRICULUM

Modules	Phase Objectives	Activities
Developing Knowledge of Game Design		
1 - 3	Collaboratively discuss, design, and develop games and understanding of core elements.	• What is a Game? • Tutorials • Game Design Schemes
Prototyping, Building and Debugging		
5-6	Prototype, build, and test games with a focus on debugging and computational thinking.	• Paper Prototyping • Game-Building • Testing & Debugging
Contributing to a Gaming Community		
6-7	Engage in peer evaluation and iterative design practices and reflection.	• Peer Playtesting • Iterative Design
8	Finalizing a workable iteration of the game and presenting to peers.	• Prepare Your Pitch • Family & Friends Game Showcase

**SNEAK
PREVIEW!**

**INTRODUCE
YOURSELF!**



Name Bulbasaur

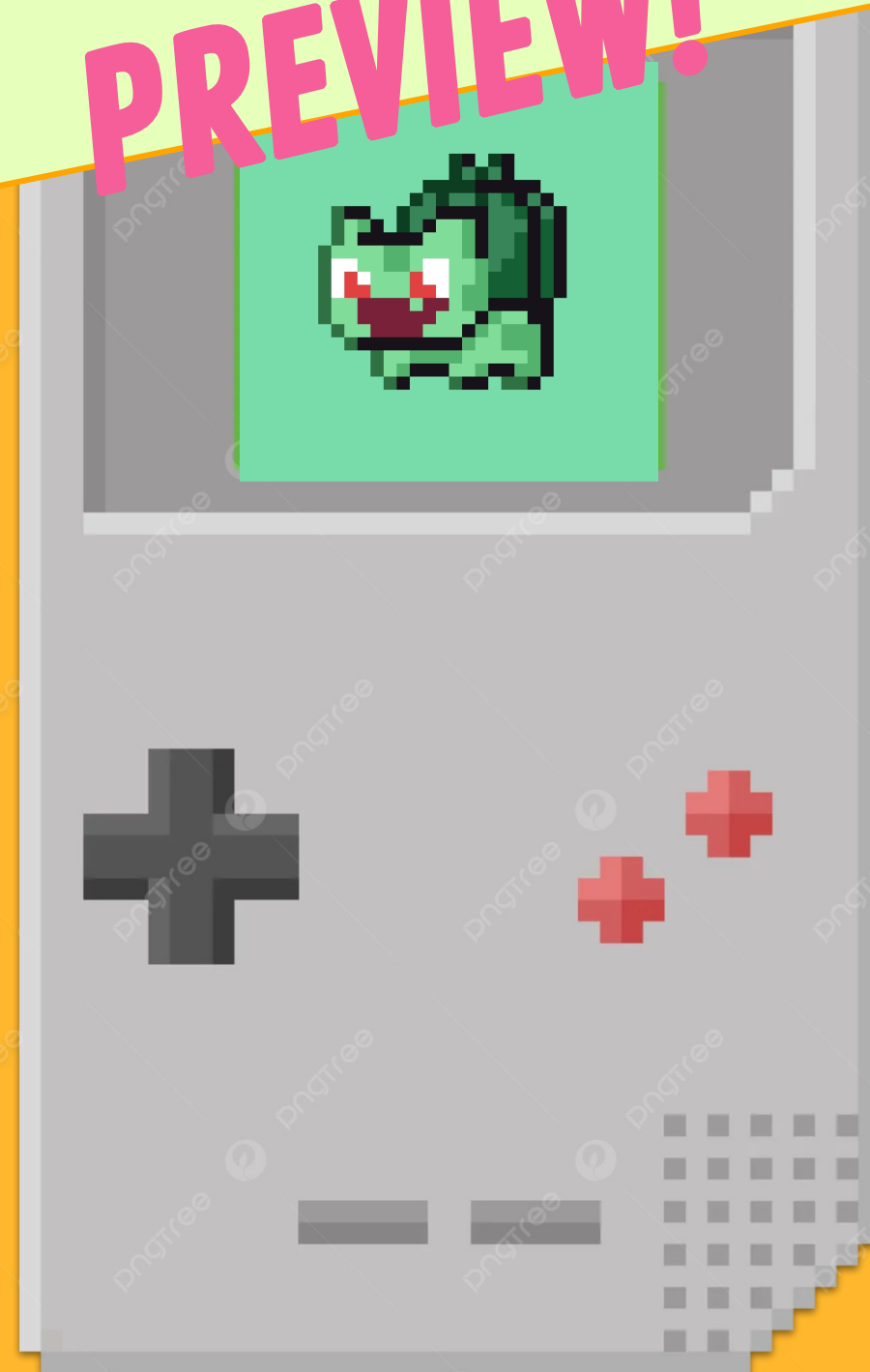
Favorite game pokemon

Why you love your favorite game

becuz i am in it :)

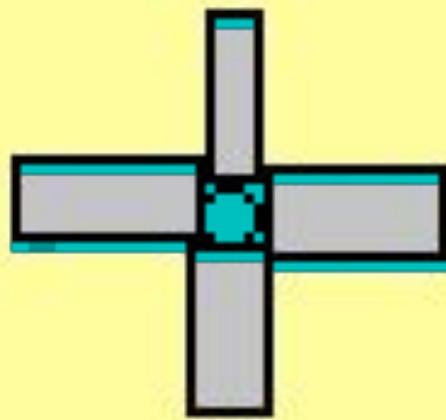
GAME BUILDER GARAGE

SNEAK
PREVIEW!



Example Game

**SNEAK
PREVIEW!**



Seedy Trouble

My name is Sam.



My game is about collecting korok seeds.

My character can walk & jump

The goal of my game is to collect all the seeds. When an item is collected my character wins
(meaning: when all the seeds are collected).

If my character touches a spiky bush then my character dies.

I will make pictures for my game. They will be
Main Character, Enemies, Collectibles

SNEAK
PREVIEW!

RQS AND DATA

METHODOLOGY

Research Questions	Quant Data	Qual Data
How can the Gaming 4 Good curriculum be implemented in informal STEM learning environments to support development of computational thinking for middle school learners?	Participant Likert Scale Reflection Data	Observations, Teacher Artifacts, Artifacts from Iterative Co-Design CAB Sessions, Youth Reflections
What is the influence of the Gaming for Good learning experience on participants' perceptions towards STEM?	Modified S-STEM Survey (Friday Institute)	Reflective Artifacts, Interviews
What is the influence of the Gaming for Good learning experience on participants' computational thinking skills?	Modified Computational Thinking Scale (CTS) (Korkmaz et al., 2017)	Game Design Documents, Game Evaluation Rubric, Videos, Observations
How does computational thinking manifest through learners' engagement with the Gaming 4 Good curriculum?		Game Design Documents, Game Evaluation Rubric, Videos, Observations



ANALYTICAL (RE)FOCUS



- Initial Findings:
 - Progression through flexible scaffolds.
 - Evidence of computing, CT, and design.
 - STEM interest and identity exploration.
- Emerging Findings:
 - Multiple forms of (dis)engagement.
 - Resistance and refusal.
 - Boundary-pushing.
 - Getting “trolled.”

Conversations in the Margins

- Participation did not always “align” with the program/tools we designed.
- We used this to iteratively redesign our curriculum and measurements.
- But we were also interested in the marginalia (Stoudt, 2016) that was produced through refusal.



RIGHTFUL PRESENCE

Kim et al. (2025) describe that rightful presence (Calabrese Barton & Tan, 2020) emerges through youth bids to reshape participation, discourse, and norms—and the extent to which those bids are recognized by adults and used to reorganize power informal environments.

Bidding might look like:

- individual or collective laughter
- resistance
- silence or subtle actions
- frustration
- humor
- critique
- proposing new directions or taking leadership

THE AMBIGUITY OF PLAY

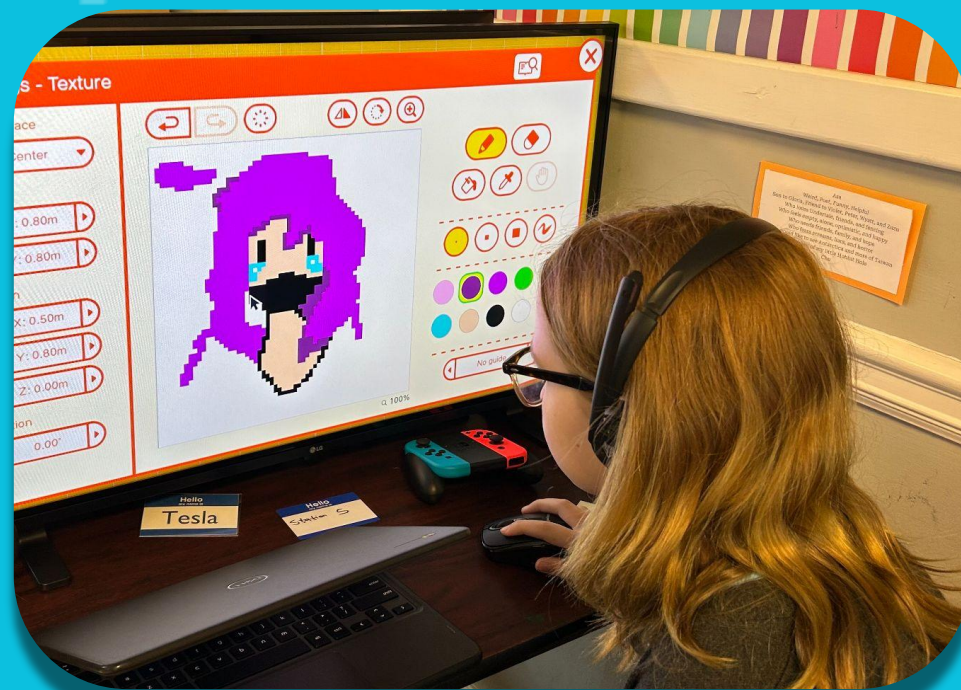
Play can be conceptualized through multiple (7), coexisting “rhetorics,” each highlighting different meanings and functions of play across contexts. (Sutton-Smith, 1997)

Most relevant to our current work:

- **Power** → play as contest, struggle, victory (e.g., strategy, competition, winning/losing)
- **Imaginary** → play as romanticism, creativity, possibility (e.g., storytelling, world-building, fantasy)
- **Self** → play as individualism, identity, expression (e.g., leisure, solitary play, extreme games)
- **Frivolity** → play as nonsense, inversion, the trickster (e.g., humor, disruption, pushing boundaries)

NOVA: THE AUTONOMOUS WORLD-BUILDER

- Rejected G4G as too simple
- Positioned herself as more advanced
- Also rejected alternative tool (Twine)
- Concept art & early design in Google Slides
- Did not produce playable builds but still engaged in CT practice and iteration
- Preferred self-directed learning
- Was very transparent with her opinions about the program and her day-to-day emotions
- Made full use of the feelings scale



LESSON 1 JOURNAL

Date: Monday

M T W T F S S

THINGS I DID TODAY

- on a roll
- tag
- made games
- Listened to an annoying ai talk and tell my what to do

TODAY I FELT

😊 😐 😞 😡 😠 😭

MY THOUGHTS ABOUT TODAY

I was hungry and tired and my Back hurt, and I Spent 7 hours - 6 hours listening to the ai's Bob, and I forgot the other one's name. Tell me what to Do.

WHAT I LEARNED TODAY

I learned how to use the world Bot.

MY RANKING OF TODAY

★ ★ ★ ★ ★

OTHER NOTES

- I Don't Like the ui Bob
- I Don't Like the game that

MILO: THE COMPETITIVE TRICKSTER

- Built platformer with hazards & restart penalties
 - Required extreme precision and “blind jumps”
 - Demonstrated high CT and computing
- Peers described game as very difficult to complete
- Responded to feedback by increasing difficulty
 - Added vertical levels with fall consequences
- Described design as “trolling” and “rage bait”
- Would regularly challenge us in his game design logs and surveys (unexpected scale responses)

Milo circled the same feeling face every week even if the star rating changed.

RAINSTORMING

Think about the type of game you'd like to create with your group. Answer the following questions to get started:

- What type of game do you enjoy playing? Platformers? Puzzle games? Racing games?
strategy
- Can you think of a story for your game? (e.g., space adventure, forest exploration, etc.)
find a sword bet battles
- How will your game challenge players and keep them engaged?
the parker + that with exit the room + to keep paly and obicak + that you
- Write down at least three ideas for the game.
*the sword is + negan
to make die by the black obstacles
character is a knight*

ADAPTING & BUILDING

Apply what you've learned from the interactive lessons to the game ideas. Reflect on these questions:

- What parts of these lessons could work well in the game ideas? Think about the notions you have learned about.
I learned how to make goal
- How can you modify or build on these ideas to fit your unique game concept(s)?
maybe did more levels
- Write down at least two features or mechanics from the lessons that you'll use or adapt. Explain how you'll incorporate them.
*no vains objects
or starting objects*

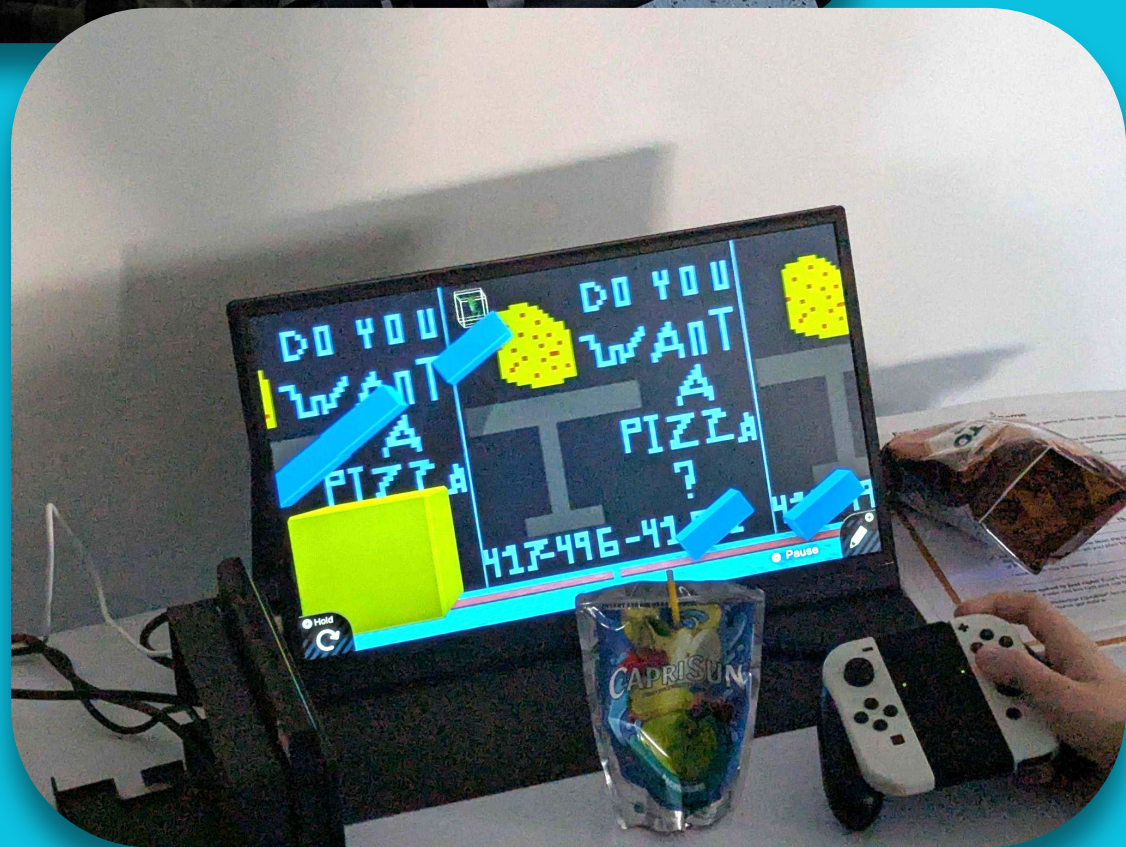
TODAY I FELT

😊 😊 😊 😊 😊 😊

MY RANKING OF TODAY

★ ★ ★ ★ ★

RYAN AND COLE: THE PLAYFUL DISRUPTORS



- Built a side-scrolling platformer with hazards
- included unintended “launch” glitch
- Glitch allowed skipping parts of the level
- Initially identified as a bug to fix
- Peers found it funny and kept triggering it
- Designers chose to keep and build around it
- Became an emergent gameplay mechanic
- Evidence of high level CT skills

LISTEN UP! (ALEMAN ET AL., 2025)

Listen up!: game jams as spaces of pedagogical rupture, counterstorytelling and youth agency

- Explored how middle school students use game design and counterstorytelling to express resistance to oppressive educational structures and create pedagogical ruptures.
- Game mechanics (e.g. limited choices, loops, and no-win scenarios) became rhetorical devices through which youth commented on the constraints of life/their experiences.
- These ruptures are small, local forms of critique “between youth and adults, power and agency, play and schooling.”

REFUSAL AS AGENCY

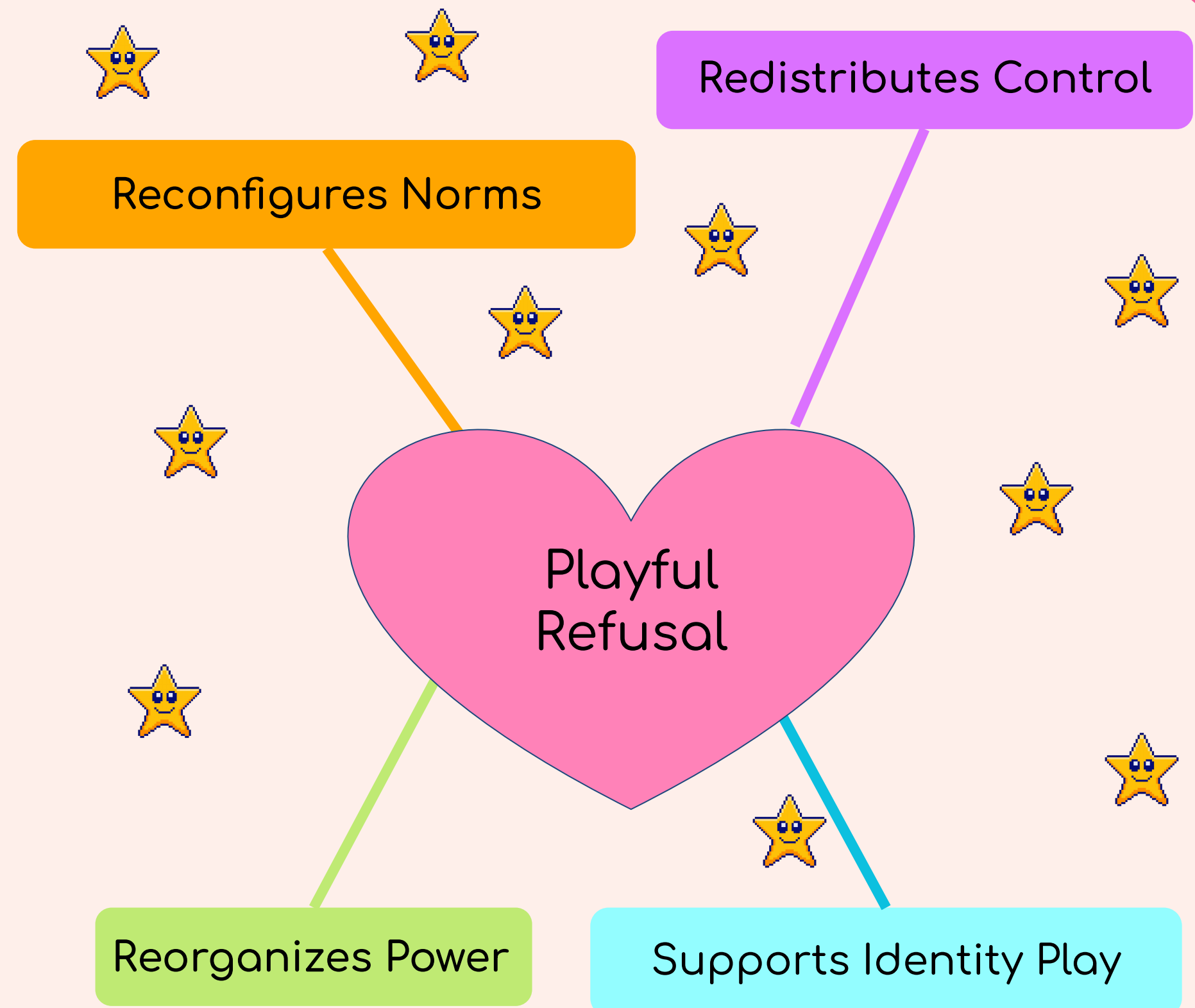
- Functions as a bid for agency, where a young person asserts control over their body, time, attention or environment (Kapp et al., 2019).
- Neurodiversity-affirming perspective, where consent, and personal agency are prioritized over compliance (Raymaker, 2020).
- Linked to autonomy and identity development in neurodivergent youth (Botha & Frost, 2020).

REFUSAL AS DISTRESS

- Refusal can also be a stress-driven response to overwhelmed nervous systems, anxiety, or sensory overload (Fisher et al., 2025).
- School-like tasks may trigger overwhelm when requiring shifts between activities that differ in interest, sensory demands, or attention (Jenson et al., 2024).
- Preparedness includes recognizing stress-based refusal and training in inclusive facilitation and transition strategies (Marsh et al., 2024).

Toward a Theory of Playful Refusal

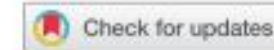
- Reconfigures norms:
 - Redefined what counted as “good” or acceptable CT or game design.
 - See (Aleman et al., 2025)
- Redistributes control:
 - Youth took ownership over decisions, tools, and the direction of their work.
 - See (Aleman et al., 2025; Kim et al., 2025)
- Reorganizes power:
 - Participants explored adult/youth control/authority within interactions.
 - See (Aleman et al., 2025; Kim et al., 2025)
- Supports identity play:
 - Identity-informed participation, choice and risk-taking.
 - See (Carlone & Mercier, 2026; Corbitt, 2024)



LESSONS LEARNED

IMPLICATIONS

- Messiness and marginalia tell important stories.
- Playfulness includes disruption, inversion, and refusal.
- Refusal is a form of communication.
- Playful Refusal can be understood as a bid to reshape participation.
- Agentic learning happens through playful refusal.
- Designing for/Expecting compliance reduces agency.
- Spaces need to feel safe enough for both play and refusal.



Call for proposals: special issue on play

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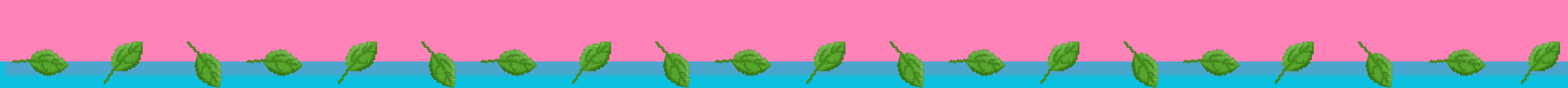
We invite proposals for a special issue of *Mind, Culture, and Activity* that explores play in all its depth and breadth – as a developmental activity, a site of resistance, a cultural and political force, and a collective, transformative practice. Across diverse global contexts, play is being reimagined and repurposed to challenge systemic injustice, to create community, and to reconfigure the boundaries between learning, activism, art, and everyday life.

This issue seeks to bring together scholars, practitioners, and playmakers who are engaging in critical and creative work on play – from early childhood classrooms to activist performance, from digital game design to grassroots organizing, from participatory design to playworlds. We aim to include both theoretical and empirical work, and we especially welcome submissions that challenge dominant paradigms or speak from historically marginalized or underrepresented perspectives.

Topics may include (but are not limited to):

- Intergenerational and intersectional approaches to play
- Play and justice: decolonial, feminist, anti-racist, and anti-capitalist interventions
- Adult play, play activism, and collective performance
- Play and the politics of learning and development
- Play and creativity in the Global South
- Digital play, games, and technological mediation
- Play and radical inclusion across ability, language, and identity
- Critical reflections on the psychology of play and its institutional forms

Thank You!

- 
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